Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
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Project Title:	Visual Orientation, Navigation, and Spatial Memory Countermeasures		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139-4301	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	03/01/2004	**End Date:**	02/28/2007
No. of Post Docs:	4	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	8
No. of Master's Candidates:	6	**No. of Bachelor's Degrees:**	7
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Harris, Laurence (York University) Liu, Andrew (Massachusetts Institute of Technology) Jenkin, Michael (York University) Aoki, Hirofumi (Massachusetts Institute of Technology)		
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	How should spacecraft designers configure interior architectural features, work areas, and the relative orientations of adjacent or docked spacecraft modules to minimize spatial disorientation and navigation problems in 0-g ? Current standards (e.g. NASA Standard 3000) offer little guidance. Can virtual reality (VR) training techniques – which astronauts currently use to plan their spacewalks - also be used to reduce the incidence of visual reorientation and inversion illusions while working inside the spacecraft? If so the incidence of space sickness should be reduced, and crews will be less disoriented when transiting modules with differently oriented visual verticals. Is a fully immersive VR system needed for such training, or could simpler, portable training systems be used? Can individual performance on operationally relevant 3D orientation, navigation, and teleoperation tasks be predicted based on simple tests of individual mental rotation and perspective taking skills? What is the best way to assess and control the direction of the perceptual vertical in an environment where there is no gravitational “down”? Can head movement contingent instability of the perceived visual world (“oscillopsia”), experienced by most returning astronauts, be quantified using visual feedback techniques? Answers can be directly applied to the design of the NASA CEV, Lunar Surface Access Module and eventually the Mars Transfer Hab interiors, to the physical arrangement of ground simulators and to the development of VR based techniques for preflight orientation and navigation training for astronauts. The goal of this multi-institutional neurovestibular project is to develop four types of design, assessment, training, and procedural countermeasures: a) Evidence-based spacecraft architecture and work area design standards. b) Methods for quantitative assessment of inflight and postflight oscillopsia. c) Preflight visual orientation training techniques to reduce disorientation and improve inflight emergency egress. d) Teleoperation procedure and training improvements based on crewmember spatial skills.
Task Description:	Our specific aims are: 1) To quantify how environmental geometric frame and object polarity cues determine human visual orientation, to support engineering and design of spacecraft work areas. 2) To develop reliable means for quantifying head-movement-contingent oscillopsia. 3) To determine whether preflight virtual reality techniques can improve astronaut 3D spatial memory and navigation abilities by reducing direction vertigo, and teaching ISS configuration and emergency egress routes. 4) To improve astronaut teleoperation performance by taking into account the mental object rotation and perspective taking abilities of individuals while training and during operations. Inflight spatial disorientation, spatial memory, navigation and teleoperation problems, and oscillopsia during re-entry and after landing have been identified as neurovestibular risks by Shuttle, Mir and ISS astronauts, NASA’s Critical Path Roadmap, the Neurovestibular Adaptation Team Strategic Plan, and a National Academy of Sciences committee report. NRA 03-OBPR-04 solicits research to determine what spacecraft architectures, interior visual cues, and preflight orientation training techniques will minimize inflight disorientation. The project utilizes the unique virtual reality research capabilities at York and MIT. Six sets of experiments are being conducted: 1) Measuring the effect of environmental geometry (frame) cues using psychophysical techniques (York). 2) Assessing the influence of polarized objects on self-orientation perception using psychophysical judgments (York). 3) Assessing the extent and pattern of head-contingent oscillopsia and visual motion (York). 4) Effect of training module orientation on inflight direction vertigo (MIT). 5) Influence of relative body orientation in preflight visual orientation and egress training (MIT). 6) Correlation of spatial abilities with simulated space station remote manipulator training performance(MIT).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results support the development of neurovestibular countermeasures for spatial orientation, navigation, and spatial memory difficulties among astronauts, and the design of future vehicles, including the CEV, Lunar Surface Access Module and later the Mars Transit Hab. Our results also pertain to human health on Earth, for example: origins and assessment of oscillopsia, and disorientation, spatial memory and navigation problems in vestibular, Alzheimer's and Multiple Sclerosis patients. Our results also pertain to the interior design of buildings to reduce disorientation by providing strong visual cues for orientation in both the vertical (to reduce falls in the elderly on stairs) and gravitational horizontal planes (e.g. the origins of “wrong door” phenomena in buildings and “geographic disorientation” in cities, and among sport orienteers, and in the design of visual cueing systems for civil and military flight simulators). To the extent that disorientation is reduced, motion sickness will also be alleviated.
Task Progress:	At York, experiments in a tumbled room and in parabolic flight demonstrated that the direction of the perceptual vertical can be reliably assessed (and mathematically modeled) by having subjects distinguish the identity of an orientationally ambiguous character (e.g. “p” vs. “d”), rather than by manually rotating a visual target or manual rod, since the latter two methods require the subject to make a judgment of the direction of the perceived gravitational vertical – something that many cannot do in weightlessness. Results from a second experiment suggest that the strength of the “levitation” visual reorientation illusion depends on scene content (scene viewed), rather than geometric field of view (view seen). Experiments in the York 6-walled immersive virtual environment (IVY) manipulating the floor/ceiling aspect ratio of simple frame cue interiors demonstrate that the surface perceived to be "the floor" depends on aspect ratio in a predictable, quantifiable manner. This year we also began experiments to see if head movement contingent instability of the perceived visual world (“oscillopsia”) experienced by most returning astronauts could be quantified using visual feedback techniques. As a surrogate stimulus for an astronaut’s return to Earth’s gravity we are using vestibular coriolis illusions produced when our normal subjects make out-of-rotation-plane head movements in a rotating chair. At MIT, our studies of how subjects learn to judge the relative orientation of two docked modules with locally consistent but incongruently aligned interior visual verticals have shown that subjects naturally tend to remember each module in a visually upright orientation. Individual performance correlates with mental rotation and perspective taking skills, and also depends on relative body orientation, and the relative orientation of the modules. Pilot tests suggest that it is the number of successive mental rotations required about principal environmental axes that determines orientation difficulty, and may explain the docked module visualization difficulties previously reported by Apollo, Mir and ISS astronauts. This year we also developed an immersive VR 3D navigation simulation based on an ISS emergency egress task, and have studied the effect of training in a locally vs globally “upright” configuration, with and without smoke obscuration. Most subjects learn quickly, but performance correlated with individual 3D spatial skills. We are currently comparing performance of subjects trained with a non-immersive (laptop) based version of the task. If subjects can learn the 3D station configuration using a simple laptop based training technique, it will greatly facilitate configuration refresher training both at home and onboard the spacecraft.

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